

Design and Implementation of a Context-Aware Smart Fan Controller Using Gesture Interaction, Thermal Hysteresis, and Occupancy-Aware Control

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Abstract

Desk fan systems typically rely on manual control or single-threshold automation, often resulting in wasted energy and poor responsiveness. This paper presents a context-aware embedded fan controller that integrates non-contact gesture control, occupancy sensing, thermal regulation, and schedule-aware operation within single microcontroller architecture. Two key contributions are proposed: (1) a triple-layer gesture-filtering algorithm incorporating majority voting, lockout timing, and reversal-detection mechanisms to suppress false-positive gesture detections; and (2) a hysteresis-based thermal-regulation state machine with a deadband buffer that eliminates actuator hunting near temperature thresholds. These features are integrated within a deterministic finite-state-machine framework governing occupancy-aware and schedule-aware control decisions. Experimental evaluation across 70 gesture trials achieved a recognition accuracy of 97.1%, while a 30-minute static-environment test produced a 0% false-positive rate. Hardware-in-the-loop validation further confirmed stable thermal-state transitions without boundary oscillation and verified correct priority-based conflict resolution. The proposed approach provides a low-complexity, microcontroller-feasible solution for reliable and energy-efficient smart fan control.

Keywords: Gesture filtering, hysteresis control, embedded systems, finite state machine, context-aware automation, microcontroller, energy efficiency

1. Introduction

Desk and small-room fans typically operate through manual control or single-threshold automation. Manual control often results in unnecessary energy consumption when fans continue operating in unoccupied spaces, whereas threshold-based automation can cause actuator hunting, characterized by repeated speed switching near temperature boundaries, leading to mechanical wear and inconsistent user experience. Another limitation of existing smart-fan systems is the lack of robust input filtering in gesture-based interfaces. Low-cost proximity sensors are susceptible to false-positive detections caused by incidental motion and environmental disturbances, which may result in unintended fan-state changes.

This paper addresses these challenges through a unified embedded control architecture that combines gesture control, occupancy sensing, thermal regulation, and schedule-aware operation. The proposed system employs deterministic finite-state-machine logic and lightweight signal-processing techniques without requiring cloud connectivity, wireless communication, or computationally intensive algorithms.

The primary contributions of this work are fourfold. First, a triple-layer gesture-filtering mechanism incorporating majority voting, lockout timing, and reversal-detection logic is developed to suppress false-positive gesture detections without additional hardware. Second, a hysteresis-based thermal state machine with a deadband buffer is implemented to eliminate oscillatory speed switching near thermal thresholds. Third, an occupancy-aware and schedule-aware control framework is introduced to restrict fan operation to relevant environmental and temporal conditions. Finally, a hierarchical priority-resolution mechanism is established to resolve conflicts among scheduling, occupancy status, thermal regulation, and gesture-based user commands.

2. Related Work

Previous research on smart fan automation can be broadly categorized into three primary domains: gesture control, occupancy-aware automation, and temperature-based regulation. In addition, time-based scheduling is frequently incorporated as a practical energy-management mechanism in embedded automation systems.

In the area of gesture control, wearable gyroscope-based approaches provide intuitive user interaction but are susceptible to sensor drift over extended operation, reducing long-term reliability [1]. Multimodal systems combining voice and gesture inputs improve interaction flexibility but increase implementation complexity and computational overhead [2]. Vision-based gesture recognition approaches employing deep-learning techniques such as YOLOv2 achieve high recognition performance; however, their processing requirements make them unsuitable for low-cost microcontroller-based embedded systems [3]. While these approaches focus primarily on gesture recognition accuracy, they generally lack dedicated mechanisms for suppressing false-positive gesture activations.

Occupancy-aware automation has been widely implemented using Passive Infrared (PIR) sensors to reduce energy consumption by disabling appliances in unoccupied spaces [4]. Although effective for motion-based occupancy detection, PIR-based systems may experience false negatives when occupants remain stationary and false positives due to thermal disturbances. Furthermore, most reported systems do not incorporate time-based operational constraints or user-controlled override mechanisms.

Temperature-based fan-control systems commonly employ threshold-driven Pulse Width Modulation (PWM) techniques to regulate fan speed according to ambient temperature [5]. While these methods are computationally efficient, they remain vulnerable to oscillatory switching behavior near threshold boundaries, commonly referred to as actuator hunting. More advanced approaches based on fuzzy-logic control can reduce this effect but introduce additional computational complexity that may be unnecessary for desk-scale cooling applications.

Time-based scheduling using Real-Time Clock (RTC) modules provides a simple and reliable mechanism for restricting appliance operation to predefined periods [6]. Such scheduling can improve energy efficiency by preventing unnecessary operation outside active hours. However, when used independently, scheduling mechanisms cannot respond to unexpected occupancy changes or dynamic environmental conditions.

Despite substantial research in these individual domains, no reviewed system simultaneously integrates gesture control, occupancy awareness, thermal regulation, and time-based scheduling within a unified embedded architecture featuring explicit conflict resolution. Moreover, existing studies generally do not implement dedicated mechanisms to suppress gesture false positives or eliminate thermal-boundary oscillation. This work addresses these limitations through a deterministic finite-state-machine architecture incorporating a triple-layer gesture-filtering mechanism, hysteresis-based thermal regulation, occupancy-aware control, and hierarchical priority resolution. A comparison of representative smart fan automation approaches is presented in Table 1.

Table 1. Comparison of Representative Smart Fan Automation Approaches

Ref	Approach	Sensing Modality	Primary Limitation
[1]	Gyroscope-based gesture control	MPU-6050 (Wearable)	Susceptible to sensor drift; no false-positive filtering
[2]	Voice and gesture multimodal control	Microphone + Gesture Sensor	Increased system complexity and processing overhead
[3]	Vision-based intelligent fan control (YOLOv2)	Camera + Embedded AI Processor	High computational requirements
[4]	PIR-based occupancy automation	PIR Sensor	Limited stationary-user detection; no scheduling support
[5]	Temperature-based PWM fan control	DHT11 Temperature Sensor	Thermal-boundary oscillation due to fixed thresholds
[6]	RTC-based time scheduling system	DS3231 RTC Module	Provides scheduling only; lacks occupancy and environmental awareness

As shown in Table 1, existing approaches address individual functionalities, while the proposed system combines gesture control, occupancy awareness, thermal regulation, and scheduling within a single embedded architecture..

3. System Architecture and Methodology

3.1 Hardware Architecture

The proposed system is implemented using a single ESP32 microcontroller responsible for sensing, decision-making, and actuation. The sensing layer consists of a DHT11 temperature sensor for thermal monitoring, an HC-SR501 PIR sensor for occupancy detection, an APDS-9960 gesture sensor for directional swipe recognition, and a DS3231 Real-Time Clock (RTC) module for schedule-aware operation. The actuation layer comprises an L298N motor driver controlling a PWM-driven DC fan. User feedback is provided through a 16×2 I2C LCD display.

The system employs I2C and single-wire communication interfaces to minimize GPIO utilization and wiring complexity. All control decisions are executed locally on the ESP32 without wireless communication or cloud dependency, ensuring deterministic operation and low-latency response. The overall hardware architecture of the proposed system is illustrated in Figure 1.

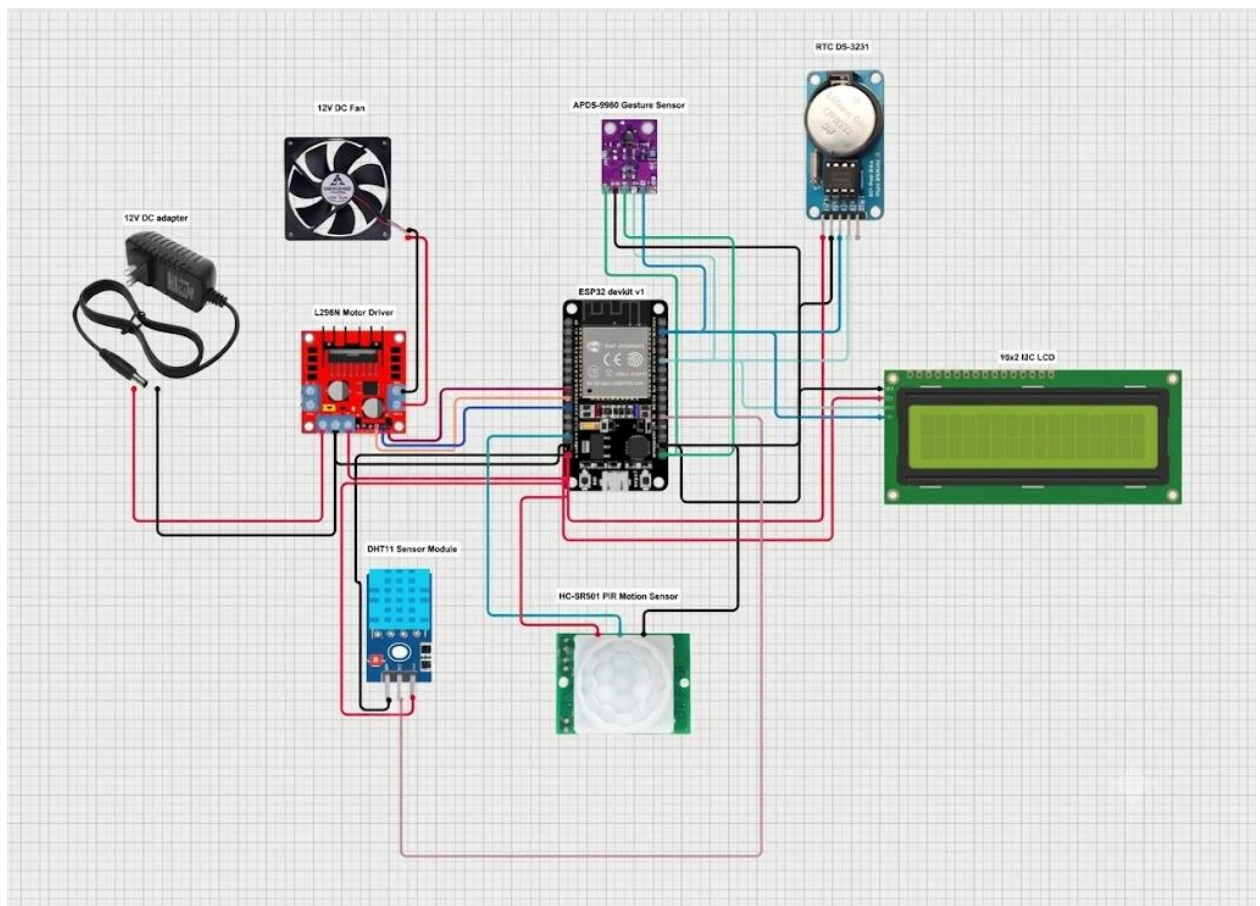


Figure 1. Overall Hardware Architecture / Circuit Diagram

The architecture consists of four sensing modules, an ESP32 processing unit, and a PWM-driven actuation layer. All sensing, decision-making, and control functions are executed locally on the ESP32 without cloud connectivity. The developed hardware prototype used for experimental validation is shown in Figure 2.

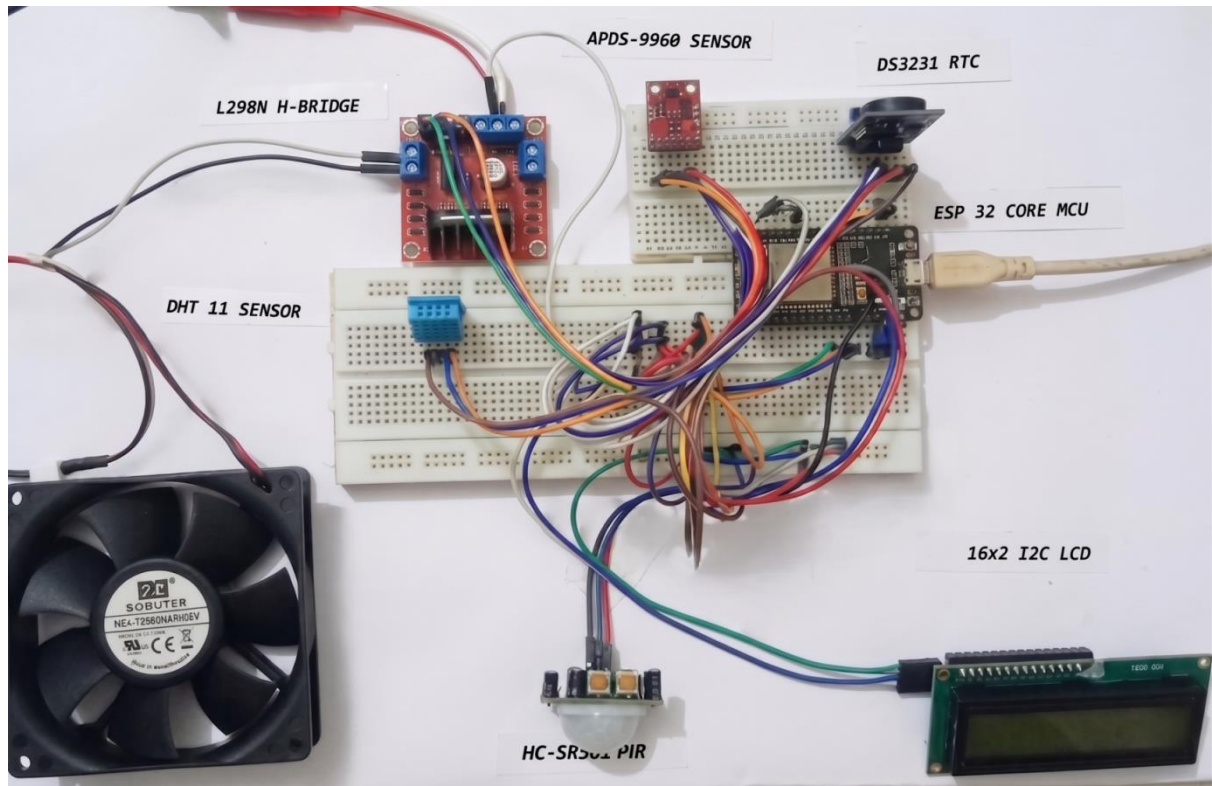


Figure 2. Developed Hardware Prototype.

The prototype integrates the ESP32 controller, DHT11 temperature sensor, PIR occupancy sensor, APDS-9960 gesture sensor, DS3231 RTC module, LCD interface, and motor-driver circuitry into a single experimental platform.

3.2 Gesture Filtering

Raw gesture outputs generated by the APDS-9960 sensor are susceptible to noise, environmental disturbances, and unintended user movements. To improve reliability, a triple-layer filtering pipeline was implemented. The overall decision-making sequence is illustrated in Figure 3.

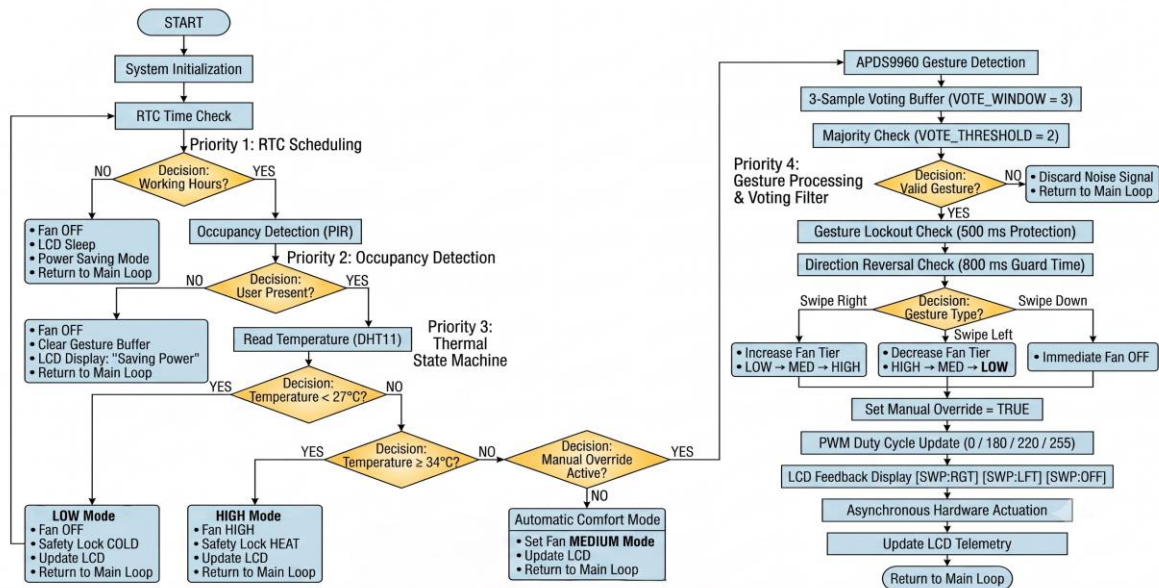


Figure 3. Firmware execution flowchart illustrating scheduling, occupancy detection, thermal regulation, gesture filtering, and fan-speed control.

The flowchart illustrates the interaction among scheduling, occupancy detection, thermal regulation, gesture filtering, and fan-speed control within the deterministic finite-state-machine architecture.

The first stage utilizes the sensor's hardware-native gesture validity flag to reject incomplete or invalid gesture events without introducing blocking delays that could affect PWM timing or overall system responsiveness.

The second stage applies a statistical majority-voting mechanism. A gesture is accepted only when at least two of the three most recent observations agree, thereby suppressing isolated sensor anomalies and transient disturbances.

The third stage introduces temporal filtering through a 500 ms lockout interval and an 800 ms reversal-detection window. The lockout interval prevents multiple triggers from a single swipe event, while the reversal-detection mechanism suppresses unintended opposite-direction gestures that may occur immediately after a valid swipe.

A gesture command is therefore accepted only when sufficient agreement exists among recent observations and when the temporal filtering conditions are satisfied. This layered approach reduces the influence of sensor noise, improves recognition reliability, and prevents unintended fan-state changes without introducing significant computational overhead.

3.3 Thermal Hysteresis

Fan-speed regulation is governed by three operating states: LOW Mode, MEDIUM Mode, and HIGH Mode. To prevent oscillatory switching near threshold temperatures, a hysteresis mechanism with a $\pm 1^{\circ}\text{C}$ deadband was implemented.

LOW Mode is entered when temperature falls below 27°C and is exited only after temperature rises above 28°C . Similarly, HIGH Mode is entered when temperature reaches or exceeds 34°C and is exited only after temperature falls below 33°C . The resulting separation between entry and exit thresholds eliminates rapid state transitions caused by minor sensor fluctuations.

Gesture control is conditionally enabled based on the current thermal state. Gesture inputs are ignored in LOW Mode (Cold Lock) and HIGH Mode (Hot Lock) and are enabled only in MEDIUM Mode, where user control does not conflict with thermal-regulation objectives.

To ensure reliable sensor operation, the DHT11 temperature sensor is polled at 2000 ms intervals. The thermal-regulation parameters implemented in the proposed controller are summarized in Table 2.

Table 2. Thermal Regulation Parameters

Parameter	Value	Description
LOW_TEMP_THRESHOLD	27°C	Entry threshold for LOW Mode
LOW_EXIT_THRESHOLD	28°C	Exit threshold for LOW Mode ($+1^{\circ}\text{C}$ hysteresis)
HIGH_TEMP_THRESHOLD	34°C	Entry threshold for HIGH Mode
HIGH_EXIT_THRESHOLD	33°C	Exit threshold for HIGH Mode (-1°C hysteresis)
DHT_INTERVAL_MS	2000 ms	Temperature sensor polling interval

These parameters establish the entry and exit thresholds used to implement hysteresis-based temperature regulation. The thermal-state transition mechanism implemented in the controller is illustrated in Figure 4.

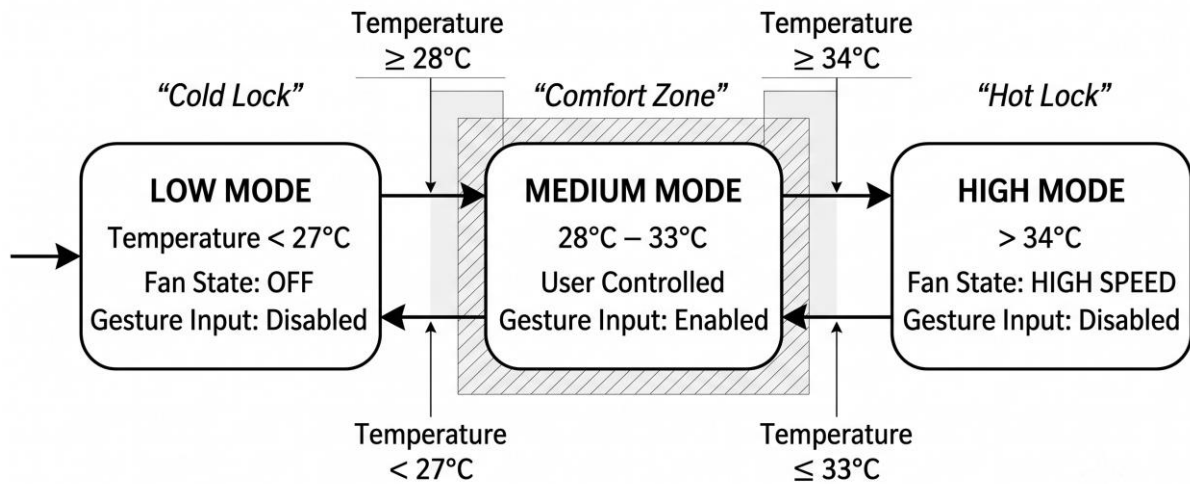


Figure 4. Thermal Hysteresis State Diagram

The state diagram demonstrates how separate entry and exit thresholds prevent rapid mode switching near temperature boundaries.

3.4 Occupancy and Scheduling

Occupancy awareness is achieved using an HC-SR501 PIR sensor. If no motion is detected for five consecutive minutes (300,000 ms), the system enters a VACANT state and forces the fan OFF to reduce unnecessary energy consumption. A `lastPresenceState` tracking mechanism prevents redundant state updates when occupancy conditions remain unchanged.

Time-based operation is enforced using the DS3231 RTC module. The system is configured to operate only between 11:00 and 23:59. Outside this interval, fan operation is disabled and the LCD backlight is switched off to minimize power consumption.

3.5 Priority Hierarchy

System decisions are resolved through a four-level hierarchical priority structure:

P1 RTC Schedule > P2 Occupancy Detection > P3 Thermal State Machine > P4 Gesture Control

The first two layers determine whether the system should operate, while the latter two determine how the system operates once active. This hierarchy ensures that safety and energy-efficiency constraints always take precedence over user convenience. The hierarchical control structure implemented in the firmware is summarized in Table 3.

Table 3. Control Priority Hierarchy

Priority	Layer	Condition	Resulting Behavior	Overrides
P1	RTC Schedule	Outside 11:00–23:59 window	Fan disabled; LCD backlight OFF	All lower layers
P2	Occupancy Detection	No motion detected for ≥ 5 minutes	Forced fan OFF	Thermal regulation and gesture control
P3	Thermal Regulation	LOW Mode or HIGH Mode	Fan forced OFF (LOW Mode) or HIGH (HIGH Mode); gesture ignored	Gesture control
P4	Gesture Control	MEDIUM Mode	User-driven fan speed adjustment permitted	None

The hierarchy ensures that schedule and occupancy constraints always take precedence over thermal regulation and user-generated gesture commands. The interaction among the control layers is illustrated in Figure 5.

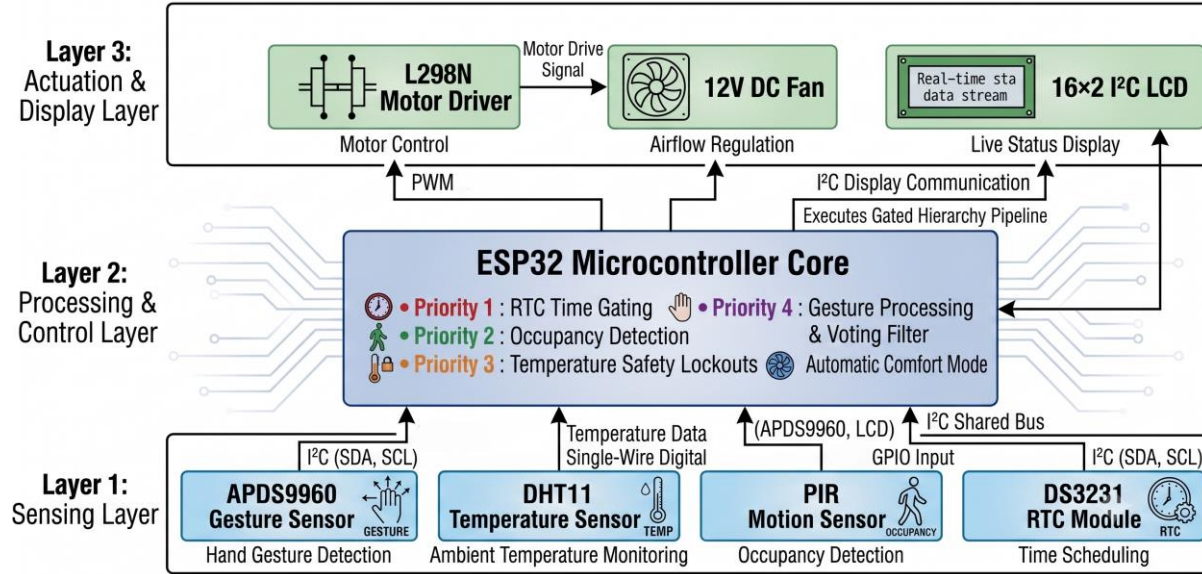


Figure 5. Priority Resolution Hierarchy.

The priority model guarantees deterministic behavior when multiple control layers simultaneously request conflicting actions.

4. Results and Discussion

4.1 Experimental Setup

System validation was conducted using a hardware-in-the-loop (HIL) testing methodology. The complete assembled platform, consisting of the ESP32 microcontroller, DHT11 temperature sensor, HC-SR501 PIR sensor, APDS-9960 gesture sensor, DS3231 RTC module, L298N motor driver, and 16x2 I2C LCD, was subjected to controlled input conditions while system behavior was observed through serial monitor output, LCD status indications, and physical actuator response. This approach was selected over simulation to ensure that timing-sensitive behaviors, including gesture processing, thermal-state transitions, occupancy timeout handling, and schedule enforcement, were validated under real operating conditions.

Validation consisted of three primary stages: (i) controlled gesture-recognition testing, (ii) static-environment false-positive evaluation, and (iii) functional validation of thermal regulation, occupancy management, scheduling logic, and priority resolution.

4.2 Gesture Recognition Accuracy

To evaluate the effectiveness of the proposed gesture filtering architecture, a total of 70 controlled gesture trials were conducted across three directional commands: LEFT (29 trials), RIGHT (20 trials), and DOWN (21 trials). Each trial consisted of a deliberate swipe performed within the normal operating range of the APDS-9960 sensor, and the recognized firmware output was compared against the intended gesture.

Of the 70 trials performed, 68 were correctly recognized, resulting in an overall gesture recognition accuracy of 97.1%.

The two unsuccessful trials did not produce unintended actuator behavior. Instead, they corresponded to instances in which the physical gesture was not fully captured by the sensor hardware and therefore failed to generate a valid gesture event. Consequently, the observed discrepancies are attributed to sensor-level acquisition limitations rather than failures of the filtering architecture itself.

4.3 False-Positive Evaluation

A primary objective of the gesture filtering mechanism is the suppression of unintended gesture activations in the absence of user interaction. To evaluate this capability, the APDS-9960 sensor was left undisturbed in a static indoor environment for a continuous observation period of 30 minutes while the firmware remained active.

No false-positive detections were observed during the entire observation period, corresponding to a 0% false-positive rate for the tested duration.

This result demonstrates that ambient disturbances and environmental noise were insufficient to satisfy the voting and temporal-filtering requirements necessary for a valid gesture command. Although the result is specific to the tested environment and duration, it confirms that the filtering architecture effectively suppresses spurious activations under normal operating conditions.

4.4 Thermal Hysteresis Validation

Tests T1–T6 were designed to validate the hysteresis-based thermal regulation strategy described in Section 3.3.

The following transition sequence was observed during controlled temperature variation:

26.8°C → Fan OFF

27.5°C → Fan OFF

28.0°C → MEDIUM Mode

34.1°C → HIGH Mode

33.5°C → HIGH Mode

33.0°C → MEDIUM Mode

The results confirm the intended asymmetric hysteresis behavior. The system remained in LOW Mode until the temperature reached the defined exit threshold of 28.0°C and remained in HIGH Mode until the temperature decreased to the defined exit threshold of 33.0°C.

The intermediate measurements at 27.5°C and 33.5°C demonstrate that the controller maintained its existing operating state despite temperature fluctuations near threshold boundaries. This behavior verifies that the implemented $\pm 1^\circ\text{C}$ hysteresis band successfully prevents repetitive state transitions and eliminates the oscillatory switching commonly observed in threshold-based control systems.

4.5 Functional Validation Matrix

The complete validation campaign conducted on the proposed system is summarized in Table 4.

Table 4. Experimental Validation Matrix

Test ID	Feature Under Test	Test Condition	Expected Behavior	Actual Behavior	Result
T1	Cold Lock Activation	Temperature = 26.8°C	Fan OFF	Fan OFF	PASS
T2	Low-Side Hysteresis	Temperature increased from 26.8°C to 27.5°C	Remain in LOW Mode (Fan OFF)	Remained in LOW Mode	PASS
T3	Low Exit Threshold	Temperature reaches 28.0°C	Transition to MEDIUM Mode	Transition successful	PASS
T4	Hot Lock Activation	Temperature = 34.1°C	Fan HIGH	Fan HIGH	PASS
T5	High-Side Hysteresis	Temperature decreased from 34.1°C to 33.5°C	Remain in HIGH Mode	Remained in HIGH Mode	PASS
T6	High Exit Threshold	Temperature reaches 33.0°C	Transition to MEDIUM Mode	Transition successful	PASS
T7	Gesture Enable Validation	Temperature within MEDIUM Mode	Gesture control enabled	Gesture control enabled	PASS
T8	Gesture Lock Validation	LOW Mode / HIGH Mode	Gesture input ignored	Gesture input ignored	PASS
T9	Occupancy Timeout Validation	No motion detected for 5 minutes	Fan OFF	Fan OFF	PASS
T10	RTC Schedule Validation	Outside operating schedule	Fan disabled; LCD backlight OFF	Expected behavior observed	PASS
T11	Occupancy Recovery Validation	Motion detected after VACANT state	Resume operation	Operation resumed	PASS
T12	Integrated Priority Validation	Simultaneous thermal and gesture events	Thermal regulation overrides gesture control	Expected hierarchy observed	PASS

All validation scenarios produced outcomes consistent with the intended firmware behavior, confirming correct implementation of the proposed control architecture.

All twelve validation tests produced outcomes consistent with the firmware design specifications, with no deviations observed during testing.

Representative LCD feedback screens observed during different operating conditions are shown in Figure 6. The display provides real-time status information, including thermal operating mode, gesture commands, occupancy state, and fan-speed indication.

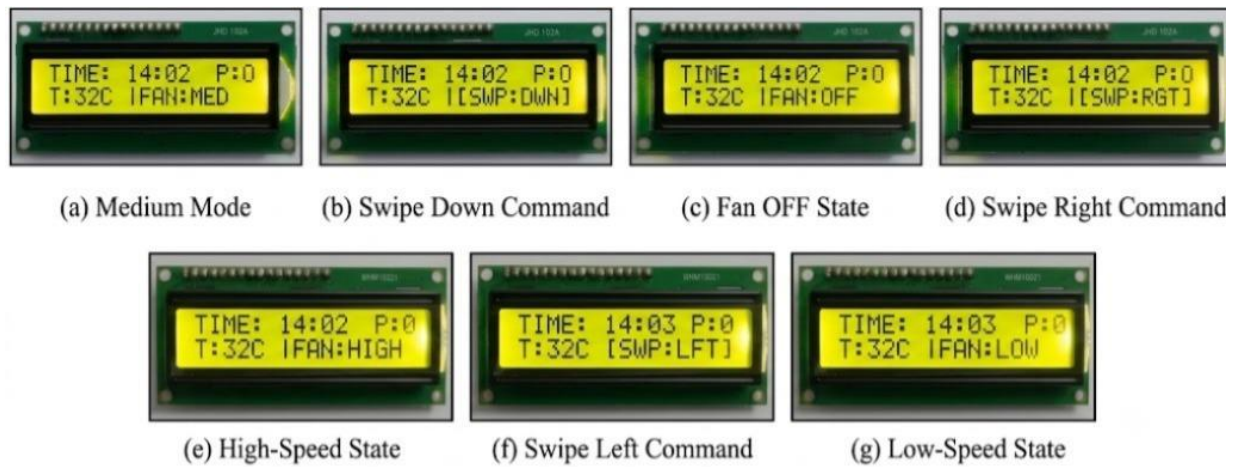


Figure 6. LCD Feedback During Different Operating States

The LCD interface provides real-time feedback regarding occupancy status, thermal state, and fan-speed selection, thereby improving system transparency and user interaction.

4.6 Discussion

The experimental results validate the primary design objectives of the proposed system. Gesture testing demonstrated that the triple-layer filtering architecture maintained high recognition accuracy while effectively suppressing unintended activations. The absence of false-positive detections during the static-environment evaluation indicates that the majority-voting and temporal-filtering mechanisms provide robust rejection of sensor noise and environmental disturbances.

Thermal hysteresis testing confirmed stable operation near temperature boundaries, eliminating oscillatory state transitions commonly observed in threshold-based controllers. Furthermore, successful completion of all functional validation tests verified correct implementation of occupancy management, RTC-based scheduling, gesture gating, and hierarchical priority resolution. In particular, Test T12 confirmed that the priority hierarchy behaves deterministically when multiple control layers simultaneously request conflicting actions.

Response latency was not instrumented during the present study and is therefore identified as a direction for future work.

Overall, the results demonstrate that layered context-aware firmware logic, implemented using deterministic finite-state machines and lightweight filtering techniques, can provide reliable and robust smart fan control without requiring machine-learning models, cloud connectivity, or computationally intensive control strategies.

5. Conclusion and Future Scope

5.1 Conclusion

This paper presented and experimentally validated a context-aware embedded fan controller integrating gesture control, thermal hysteresis regulation, occupancy detection, and RTC-based scheduling within a deterministic finite-state-machine architecture. The proposed triple-layer gesture-filtering mechanism achieved a gesture recognition accuracy of 97.1% (68 successful detections out of 70 trials), while no false-positive gesture activations were observed during a 30-minute static-environment evaluation. The hysteresis-based thermal-regulation strategy successfully eliminated boundary oscillation and ensured stable fan operation across varying temperature conditions.

Hardware-in-the-loop validation further confirmed the correct implementation of occupancy management, scheduling logic, gesture gating, thermal safety lockouts, and priority-based conflict resolution. The results demonstrate that reliable and energy-efficient fan control can be achieved using lightweight embedded algorithms without requiring cloud connectivity, machine-learning models, or computationally intensive processing.

5.2 Limitations

The present work has several limitations. Response latency between gesture detection and fan actuation was not instrumented or quantitatively measured. Experimental validation was conducted within a single indoor environment and over a limited testing duration. In addition, the PIR sensor is inherently unable to detect stationary occupants, which may result in occupancy misclassification despite the implemented five-minute timeout strategy. Finally, the thermal thresholds are fixed and do not adapt to individual user comfort preferences.

5.3 Future Scope

Future enhancements may focus on three key directions. First, the PIR sensor may be replaced with a millimetre-wave radar sensor such as the HLK-LD2450 to enable reliable human-presence detection, including stationary occupants. Second, on-board data logging using EEPROM or flash memory can be incorporated to record temperature trends, occupancy patterns, and user interaction data for long-term analysis. Third, the collected operational data can be utilized to develop a TinyML-based adaptive thermal-control framework capable of learning user preferences and dynamically adjusting fan behaviour. These enhancements would extend the proposed system from a rule-based controller toward an intelligent and personalized embedded automation platform while preserving its low-cost and resource-efficient design.

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